

ECS401: Unit 6 Lab Progress Test

FEEDBACK: Answers and Explanations

1. Write the header line for a method called `eat` which takes no arguments but returns a `String` result.

CORRECT ANSWER:

```
public static String eat()
```

EXPLANATION:

The header line is just the first line of the method. It starts `public static` then has the return type (here `String`) the name (`eat`) and then the arguments in parentheses. As there are no arguments nothing is put in the parentheses in this example.

Consider the following method:

```
public static void wombat (int a)
{
    String b = "1";
    System.out.println(a + " " + b);
    return;
}
```

2. What is the return type of method `wombat`?

CORRECT ANSWER:

`void`

EXPLANATION:

The return type is the type given before the name of the method in the header line. Here it is `void` which is a type with no values, meaning the method returns nothing.

3. What is the formal parameter of method `wombat`?

CORRECT ANSWER:

`a`

EXPLANATION:

The method has one formal parameter – there is one variable declared in the parentheses after the method name. The declaration is `int a` so `a` is the formal parameter. That means the method should be given one argument when called. That argument's value is passed to the method and will be stored in variable `a` for use in the method.

4. What type must arguments passed to `wombat` be?

CORRECT ANSWER:

`int`

EXPLANATION:

The formal parameter `a` is declared as an integer variable, meaning that the values passed to the method must be integers.

5. Is `wombat` a function or a procedure?

CORRECT ANSWER:

`procedure`

EXPLANATION:

Anything with a `void` return type is a procedure. It does not return a value. If any type other than `void` had been given then it would be a function.

6. Give the text that is printed to the screen when the following call is made.

```
wombat(5);
```

CORRECT ANSWER:

5 1

EXPLANATION:

The value 5 is passed as an argument. It is stored in the local variable, `a`, of `wombat`. It puts "1" in to the local variable `b`. The value in `a`, a space then the value in `b` are printed so 5 1 appears on the screen.

7. Give the text that is printed to the screen when the following call is made.

```
int a = 4;  
wombat(6);
```

CORRECT ANSWER:

6 1

EXPLANATION:

A variable `a` is set to 4 but it is a completely different variable to the parameter of `wombat` which is local to the method. They are different storage locations that happen to have the same name. The value 6 is passed to `wombat` so it is 6 that goes in `wombat`'s variable `a` and so it is a 6 that is printed.

8. Give the text that is printed to the screen when the following call is made.

```
int a = 4;  
wombat(a);
```

CORRECT ANSWER:

4 1

EXPLANATION:

This time the value in a (ie 4) is passed to the method. That 4 is stored in wombat's local variable a and so is printed out.

9. Give the text that is printed to the screen when the following call is made.

```
int a = 9;  
int b = 7;  
wombat(a);
```

CORRECT ANSWER:

9 1

EXPLANATION:

The b here is a different variable to wombat's b, so the fact that it has 7 stored in it does not affect what wombat does. 9 is stored in a and that value is passed as an argument, so 9 is stored in wombat's variable called a. It is therefore 9 that is printed.

10. Give the text that is printed to the screen when the following call is made.

```
int a = 4;  
int b = 7;  
wombat(b);
```

CORRECT ANSWER:

7 1

EXPLANATION:

a is given the value 4 but as it is not passed as an argument it does not have any bearing on what wombat does. b is given the value 7 and this value is passed to wombat. It is stored in the parameter a (wombat's version of it) and so that 7 is printed. wombat's variable b is a different local variable that is not affected by the changes, so it holds the string "1" still. That means 7 (ie a) and then 1 (ie b) are printed separated by a space.

11. Write a method called `checkNumber` that takes two arguments: an integer and a string. The string can be assumed to be a series of digits. `checkNumber` should check if the string holds the same number as the integer. It should return a boolean result giving the answer.

HINT: Use `Integer.parseInt` to convert the string so it can be compared with the integer.

CORRECT ANSWER:

Here is one possible answer that uses an if statement:

```
public static boolean checkNumber(int n, String s)
{
    int sn = Integer.parseInt(s);
    if (sn == n)
        return true;
    else
        return false;
}
```

Here is an alternative way of doing it that just returns the boolean value of the test directly:

```
public static boolean checkNumber(int n, String s)
{
    int sn = Integer.parseInt(s);
    return (sn == n);
}
```